GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Comprehensive GATE Notes: Electromagnetics (EMFT)

Paper Code: EC • Section 8: Electromagnetics (8-11 Marks)

Subject Code: EC-EMFT • **Target Exam:** GATE ECE • **Weightage:** 8-11 Marks

1. Transmission Lines, Waveguides & Antennas

- **Lossless Transmission Line:** Z_0 = sqrt(L / C). Reflection coefficient Gamma_L = (Z_L - Z_0) / (Z_L + Z_0). VSWR S = (1 + |Gamma|) / (1 - |Gamma|).
- **Quarter-Wave Line:** Z_in = Z_0^2 / Z_L. Used for impedance matching.
- **Rectangular Waveguide:** Dominant mode is TE_10 with cutoff frequency f_{c,10} = c / (2 a).

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Reflection Coefficient	**Gamma_L = (Z_L - Z_0) / (Z_L + Z_0)**	Z_L, Z_0 = Load, Char Impedance	Voltage reflection ratio at load termination
Quarter-Wave Transformer	**Z_{in} = Z_0^2 / Z_L**	Length l = lambda / 4	Transforms load impedance into inverted dual impedance
TE_10 Cutoff Frequency	**f_{c,10} = c / (2 a)**	a = Broader Waveguide Dimension	Lowest frequency that can propagate down hollow waveguide

GATE PYQ (2 Marks) - Transmission Line SWR Calculation
Question: 50 Ohm lossless line is terminated with Z_L = 100 + j50 Ohms. Find |Gamma_L| and VSWR.

Step 1: Gamma_L = (100 + j50 - 50) / (100 + j50 + 50) = (50 + j50) / (150 + j50) = (1 + j)/(3 + j).
Step 2: |Gamma_L| = sqrt(2) / sqrt(10) = 1 / sqrt(5) ≈ **0.4472**.
Step 3: VSWR S = (1 + 0.4472) / (1 - 0.4472) = 1.4472 / 0.5528 = **2.62**.

Essential GATE Exam Pitfalls & Scoring Traps

- **Important:** In waveguides, phase velocity v_p > c, group velocity v_g < c, and v_p * v_g = c^2.
- **Important:** Half-wave transmission line reproduces load identically: Z_in = Z_L.